TATA	FIR VEHICLE TESTING	12/06/2023

FIR No :FIR/Testing-Outdoor/516900/2023/6095

Part No/ Group :51695248R

Subject :Report on fluid economy evaluation of TML Signa 2821.T Cx 5LNGE (204 HP engine) OBDII BS-VI vehicle with G950 (FGR: 9.36) Transmission and 5.12 Rear axle ratio in comparison with TML Signa 2821.T BS-VI OBD-I vehicle as a Base vehicle.

Test Vehicle	Project Name :Signa 2821.T Cx 5LNGE (204 HP engine) OBDII BS-VI Vehicle Model Name : Signa 2821.T Cx 5LNGE	Enclosures : Annexure 1 - 6
	Chasis Number :PC03509022RJ00800 Engine Number :GXX103502	
FIR Status :	ESO FIR	

Role	Name	Signature
Prepared By	Palash Das	12-Jun-2023
Checked By	Swapnil Jadhav	12-Jun-2023
Approved By	Ajaykumar Jadhav	12-Jun-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

Documents : Please click here

FIR_Testing_Outdoor_51690..

Subject: -	Report on fluid economy evaluation of TML Signa 2821.T Cx 5LNGE (204 HP engine) OBD-II BS-VI vehicle with G950 (FGR: 9.36) Transmission and 5.12 Rear axle ratio in comparison with TML Signa 2821.T BS-VI OBD-I vehicle as a Base vehicle.

1.0 Conclusion:

1.1 Fluid economy in rated load condition at 40 kmph :

- Fluid economy of Signa 2821.T Cx BS-VI OBD-II vehicle was found **4.66 kmpl** which was Superior by **4.84%** wrt Signa 2821.T BS-VI OBD-I Base vehicle.

1.2 Fluid economy in rated load condition at 55 kmph :

- Fluid economy of Signa 2821.T Cx BS-VI OBD-II vehicle found **4.20 kmpl** which was superior by **3.93%** wrt Signa 2821.T BS-VI OBD-I Base vehicle.

Action by: -- Failure class: --

1.3 Fluid economy in Unladen condition at 40 kmph :

- Fluid economy of Signa 2821.T Cx BS-VI OBD-II vehicle found **8.44 kmpl** which was superior by **7.32%** wrt Signa 2821.T BS-VI OBD-I Base vehicle.

Action by: -- Failure class: --

1.4 Fluid economy in Unladen condition at 55 kmph :

- Fluid economy of Signa 2821.T Cx BS-VI OBD-II vehicle found **7.21 kmpl** which was which was Superior by **6.32%** wrt Signa 2821.T BS-VI OBD-I Base vehicle.

Action by: -- Failure class: --

1.5 Overall fluid economy status at 40 kmph (80% Laden + 20% unladen condition):

- Fluid economy of Signa 2821.T Cx BS-VI OBD-II vehicle found **5.12 kmpl** which was Superior by **5.14%** wrt Signa 2821.T BS-VI OBD-I Base vehicle.
- Signa 2821.T Cx BS-VI OBD-II meets the PALS target of “Leadership”.

1.6 Overall fluid economy status at 55 kmph (80% Laden + 20% unladen condition):

- Fluid economy of Signa 2821.T Cx BS-VI OBD-II vehicle found **4.59 kmpl** which was Superior by **4.23%** wrt Signa 2821.T BS-VI OBD-I Base vehicle.
- Signa 2821.T Cx BS-VI OBD-II meets the PALS target of “Leadership”.

Action by: -- Failure class: --

FIR Status

2.0 Result: Vide Observations

FAILURES CLASS:

A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.

B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.

C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.

D: Customer perception issues and minor irritants.

3.0 Test Vehicles: -

1. Signa 2821.T BS-VI OBD-II (ERCJ-1248)_Test vehicle

   Chassis no. - PC03509022RJ00800; Engine no. - GXX103502.

2. Signa 2821.T BS-VI OBD-I (ERCJ 1187)_Base vehicle

   Chassis no. - PC13518022RJ00784; Engine no. - GXX103780.

Input Doc Ref.: - Test Request No: - Testing-Outdoor/516900/2023/6095

DID No: – NA.

DDP No: – NA.

Enclosures:

Annexure 1: Panel chart

Annexure 2: PAT sheet

Annexure 3: Duty cycle report

Annexure 4: Test request

Annexure 5: Test plan

Annexure 6: EOL

Work Order No. :- P.22.EA.501	**Make :-** TATA Motors Limited	**T I No. – 42163**

Earlier Ref: - Nil.

4.0 Project Status:- DR3

5.0 Introduction:

As a part of RDE (OBD-II) Testing Program, TML Signa 2821.T Cx 5LNGE BS-VI OBD-II vehicle with G950 (FGR:9.36) Transmission and 5.12 Rear axle ratio was offered for fluid economy evaluation along with TML Signa 2821.T 5LNGE BS-VI OBD-I vehicle as a base vehicle. All vehicles were tested back-to-back at JSR- Ranchi-JSR route in laden and unladen conditions at 40 km/hr as well as 55 km/hr vehicle cruising speeds. This report will reveal details of this trial with the status.

6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fluid economy evaluation	TST/VT/WG/158	Data Logger

7.0 Vehicle details:

Parameter		Base Vehicle	Test vehicle
Vehicle model		Signa 2821.T (OBD-I) (ERC-1187)	Signa 2821.T (OBD-II) (ERC-1248)
A) Vehicle specification			
Engine Model		Tata 5LNGE BS-VI (200 HP) OBD-I	Tata 5LNGE BS-VI (204 HP) OBD-II
Max Power		200 HP @2200 RPM	204 HP @2200 RPM
Max Torque		850 Nm @ 1000 ~1600 rpm	850 Nm @ 1000 ~1600 rpm
GVW (kg)		28000	28000
Gear Box		Model: Tata G950, Type: DD	Model: Tata G950, Type: DD
Gear Ratios.		1st gear:9.36, 2nd gear: 5.2, 3rd gear: 3.05, 4th gear: 1.84, 5th gear: 1.33, 6th gear: 1.00	1st gear:9.36, 2nd gear: 5.2, 3rd gear: 3.05, 4th gear: 1.84, 5th gear: 1.33, 6th gear: 1.00
Axle Ratio		4.88	5.12
Tires	FA1	Size: 295/90R20, Make: JK , Type: LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	FA2	Size: 295/90R20, Make: JK , Type: LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	LA1	Size: 295/90R20, Make: JK , Type: LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	RA1	Size: 295/90R20, Make: JK , Type: LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	RA2	Size: 295/90R20, Make: JK , Type: LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
Tractive effort @ 1st gear		74736 N	78411 N
Tractive effort @ Top gear		7985 N	8377 N
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"	Type: E-Viscous fan and Size:24"

Parameter	Base Vehicle	Test vehicle
Vehicle model	Signa 2821.T (OBD-I) (ERC-1160)	Signa 2821.T (OBD-II) (ERC-1256)
B) FE Avenues _Engine Level		
Base Cal Improvement (Cal Maturity %)	As per production released	95% Cal Maturity
Low Viscous Engine Oil (TMGO-10W30)	No (15W40CK4+)	Yes (10W30)
FE Switch	Yes (3 Mode : E/B/P)	Yes (3 Mode : E/B/P)
Engine Features (EOL : Enable/Disable)	Disable	Disable
ATS Optimization	No	No
APM	No	No
Cruise Control	No	No
Gear based VAM	No	No
Intelligent mode shifter	No	No
Moving Mass Estimation	No	No
C) FE Avenues _Vehicle Level		
Driveline Optimization	NA	No
GSA (Gear Shift Advisor)	Yes	Yes (Not Tuned)
MSA (Mode Shift Advisor)	NA	No
Low CRR Tyres	Yes	Yes
Low Viscous rear axle oil (80W90)	Yes	Yes
Smart Alternator (LIN)	NA	No
THU/Unitised/Low friction bearing	Yes	Yes

8.0 Test result:

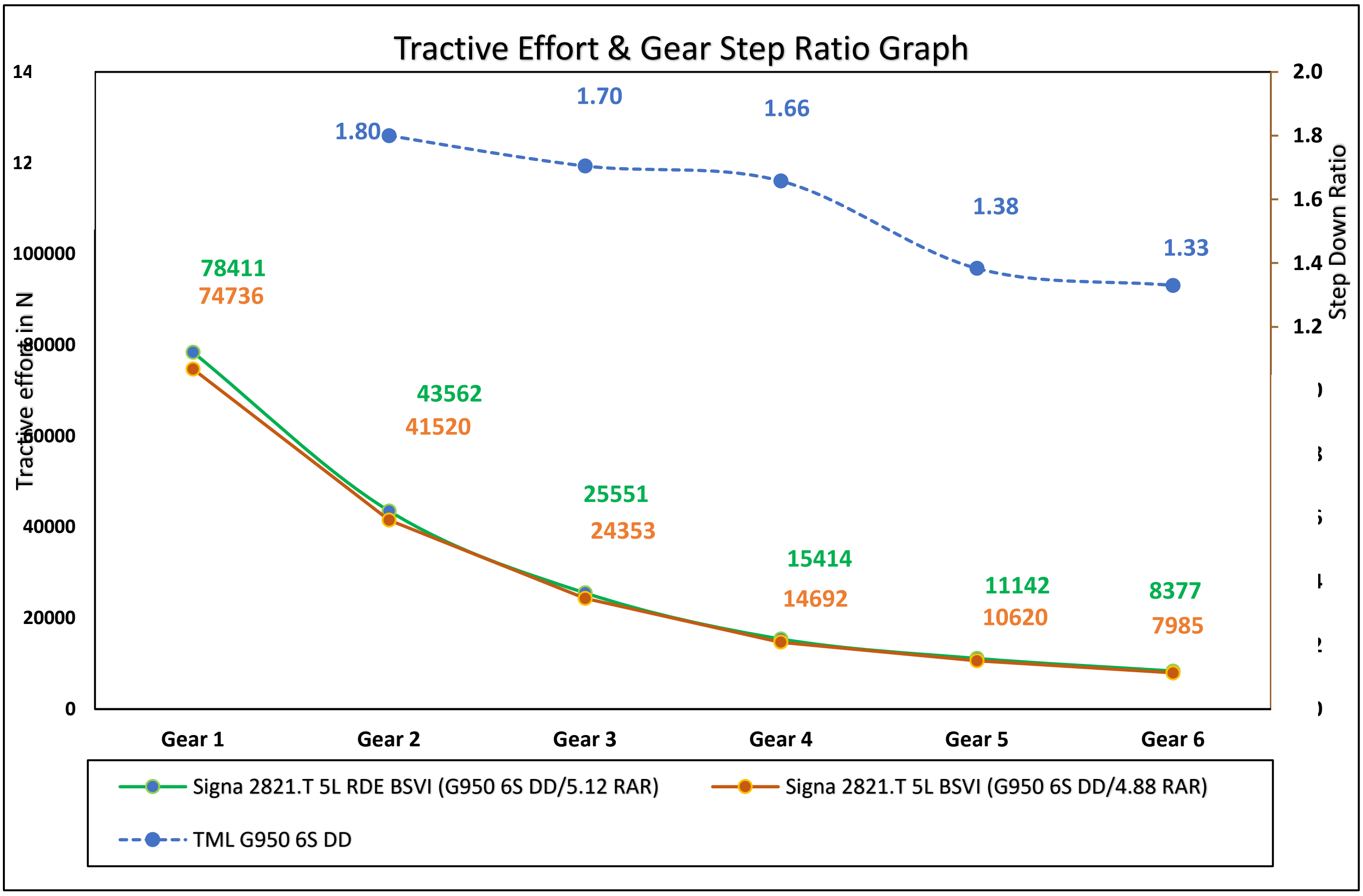
8.1 Cruising speed: 40 Kmph & 55 Kmph :

Sr. No.	Parameters		Unit	Base Vehicle : OBD-1 SIGNA 2821.T Tata 5L BSVI	Test Vehicle : OBD-2 SIGNA 2821.T Tata 5L RDE BSVI
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1187	ERC-1248
2	Vehicle GVW/GCW		kg	28000	28000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			TML 5LNGE	TML 5LNGE
	Engine Max Power			200 HP @ 2200 rpm	204 hp @ 2200 rpm
	Engine Max Torque			850 Nm @ 1000-1600 rpm	850 Nm @ 1000-1600 rpm
	Engine Calibration Maturity Level			As per production released	95%
4	Regeneration Strategy (Time ase/Soot base)			110 Hrs.	110 Hrs.
5	Gear Box (make/drive/FGR)			**TATA G-950 DD**	**TATA G-950 DD**
	Gear Ratios			1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00
6	RAR			4.88	5.12
7	Tyre Details			295/90 R 20 RIB JK low CRR	295/90R 20 RIB Bridgestone Low CRR
8	LBSC	High & Low Engine Speed Breakpoint	RPM	--	--
		Vehicle Mass Threshold	kg	--	--
	VAM	Variable Acceleration 1, 2	kmph/sec	--	--
		Variable Acceleration Speed 1, 2	kmph	--	--
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	--	--
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	100%	--
		Torque Limit Maximum Adjustment Factor	Economy	100%	--
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable
		Torque Limit	Nm	NA	NA
	USFE / Torque Table Revision (1/2/3/n)			--	--
9	Fan Details (type/make/blades)			E- 24" High Efficiency Viscous fan	E- 24" High Efficiency Viscous fan
FE Avenues (Engine Level)	Base Cal Improvement (Cal Maturity %)			As per production released	95%
	Low Viscous Engine Oil (TMGO-10W30)			No (15W40CK4+)	Yes (15W40CK4+)
	FE Switch			Yes (3 Mode : E/B/P)	Yes (3 Mode : E/B/P)
	Engine Features (EOL : Enable/Disable)			Disable	Disable
	ATS Optimization			No	No
	APM			No	No
	Cruise Control			No	No
	Gear based VAM			No	No
	Intelligent mode shifter			No	No
	Moving Mass Estimation			No	No
FE Avenues (Vehicle Level)	Driveline Optimization			NA	No
	GSA (Gear Shift Advisor)			Yes	Yes
	MSA (Mode Shift Advisor)			NA	No
	Low CRR Tyres			Yes	Yes
	Low Viscous rear axle oil (80W90)			Yes	Yes
	Smart Alternator (LIN)			NA	No
	THU/Unitised/Low friction bearing			Yes	Yes
FE Trial Details	Route			**Jamshedpur-Ranchi-Jamshedpur**	**Jamshedpur-Ranchi-Jamshedpur**

	Parameters	Unit	Base: Laden Trial @ 40 kmph		Base: Unladen Trial @ 40 kmph		Base: Laden Trial @ 55 Kmph			Base: Unladen Trial @ 55 kmph		Test: Laden Trial @ 40 kmph		Test: Unladen Trial @ 40 kmph		Test: Laden Trial @ 55 Kmph			Test: Unladen Trial @ 55 kmph	
FE Trial Details	FE Duty Cycle (Laden/Unladen)		Laden Trial @ 40 kmph		Unladen Trial @ 40 kmph		Laden Trial @ 55 Kmph			Unladen Trial @ 55 kmph		Laden Trial @ 40 kmph		Unladen Trial @ 40 kmph		Laden Trial @ 55 Kmph			Unladen Trial @ 55 kmph	
	Trial Date		05.05.2023	06.05.2023	14.05.2023	16.05.2024	20.04.2023	29.04.2023	04.05.2023	11.05.2023	13.05.2023	05.05.2023	06.05.2023	14.05.2023	16.05.2023	20.04.2023	29.04.2023	04.05.2023	11.05.2023	13.05.2023
	Weather Condition (Sunny/Rainy/Cloudy/Windy)		Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny
	Traffic Condition (Light/Average/Heavy)		Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average
	Regeneration (Yes/No)		No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No
	Fuel spent in Regen	%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	Fuel spent in SCRTM	%	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited
	Trip Distance	km	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00
	Total Diesel Quantity	Litre	53.40	53.42	31.38	28.91	59.83	57.64	59.26	35.00	35.31	50.25	50.97	28.95	27.22	58.09	54.96	56.46	32.7	33.59
	Total Urea Quantity	Litre	1.20	2.00	0.90	1.30	3.00	0.00	1.20	0.90	0.60	2.20	2.20	0.90	1.10	2.70	0.70	1.80	0.8	0.3
	Equivalent Urea Quantity : 50% Consumption	Litre	0.60	1.00	0.45	0.65	1.50	0.00	0.60	0.45	0.30	1.10	1.10	0.45	0.55	1.35	0.35	0.90	0.40	0.15
	Total Eq. Fluid Consumed	Litre	54.00	54.42	31.83	29.56	61.33	57.64	59.86	35.45	35.61	51.35	52.07	29.40	27.77	59.44	55.31	57.36	33.10	33.74
	Urea uses % wrt. Diesel	%	2%	4%	3%	4%	5%	0%	2%	3%	2%	4%	4%	3%	4%	5%	1%	3%	2%	1%
	Fuel Economy	kmpl	**4.51**	**4.51**	**7.68**	**8.34**	**4.03**	**4.18**	**4.07**	**6.89**	**6.83**	**4.80**	**4.73**	**8.32**	**8.85**	**4.15**	**4.39**	**4.27**	**7.37**	**7.17**
	Status of Fuel economy	%	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	6.27%	4.81%	8.39%	6.209%	3.00%	4.88%	4.96%	7.034%	5.12%
	Equivalent Fluid Economy	kmpl	**4.46**	**4.43**	**7.57**	**8.15**	**3.93**	**4.18**	**4.03**	**6.80**	**6.77**	**4.69**	**4.63**	**8.20**	**8.68**	**4.05**	**4.36**	**4.20**	**7.28**	**7.14**
	Status of Equivalent Fluid economy	%	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	5.16%	4.51%	8.27%	6.45%	3.18%	4.21%	4.36%	7.10%	5.5%
	Cluster Total Odo. Reading (at End of trial)	km	230	230	230	230	230	230	239	230	230	253	253	250	250	253	253	253	250	250
	Cluster Trip AFE	kmpl	4.50	4.50	8.00	8.30	3.90	4.00	4.10	7.10	6.90	5.10	5.00	8.70	9.00	4.60	4.50	4.60	7.90	7.80
	Status of Cluster Trip AFE wrt Actual fluid economy	%	0.83%	1.61%	5.66%	1.80%	-0.75%	-4.33%	1.84%	4.44%	1.95%	8.67%	8.03%	6.13%	3.71%	13.45%	3.28%	9.48%	8.50%	9.20%
Avg.kmpl Results	**Average Fuel Economy**	kmpl	4.51		8.01		4.09			6.86		4.76		8.59		4.27			7.27	
	Status of Average Fuel economy	%	Base		Base		Base			Base		5.54%		7.26%		4.29%			6.08%	
	Average Eq. Fluid Economy	kmpl	4.45		7.86		4.05			6.78		4.66		8.44		4.20			7.21	
	Status of Average Eq. Fluid economy	%	Base		Base		Base			Base		4.84%		7.32%		3.93%			6.32%	
Overall Results L+UL	**Overall Avg. Fuel Economy (kmpl) : L + UL**	80% / 20%	4.94				4.45					5.23				4.65				
	Status of Overall Avg. Fuel Economy (%)		Base				Base					5.75%				4.52%				
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL	80% / 20%	4.87				4.40					5.12				4.59				
	Status of Overall Avg. Eq. Fluid Economy (%)		Base				Base					5.14%				4.23%				

Note: Equivalent Urea consumption is taken as 50% of actual urea consumption.

8.3 Tractive Effort Comparison :



Remarks:

Signa 2821.T Cx BS-VI OBD-II (204 HP) Test vehicle tractive effort was better in all gears as compared with Signa 2821.T BS-VI OBD-I (200 HP) _ Base vehicle.

8.4 Torque Table :

8.3.1 : Base vehicle (Signa 2821.T BS-VI OBD-I (200 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gear	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	-	850	850	700	590	590	590	-	-	850
Balance Mode	-	850	850	700	700	700	700	-	-	850
Power Mode	-	850	850	850	850	850	850	-	-	850

8.3.2: Test vehicle (Signa 2821.T BS-VI OBD-II (204 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gear	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	NA	850	850	700	590	590	590	-	-	850
Balance Mode	NA	850	850	700	700	700	700	-	-	850
Power Mode	NA	850	850	850	850	850	850	-	-	850

9.0 Observation: Nil

10.0 Further Testing: Further fuel economy evaluation needs to be done with benchmark vehicle to get the PALS attribute wrt core competitor.

Annexure-1

Level 1 Attribute:	PANEL Chart	VE CVBU/FE/02
PAT- Fuel Economy	Specification:Max Power- 204 hp @ 2200 rpm Max Torque - 850 Nm 1000-1600 rpm Driveline : G-950 DD | 5.12 RAR | 295/90R 20 Tyres | 3 Mode FE Switch | Torque Factor: NA EOL Parameters : RSG: 80 kmph	Panel Chart No: VE CVBU/ HCV/ SIGNA 2821.T 5L Cx BSVI RDE/ Cx / FE/02

Programme / MY : SIGNA 2821.T Cx 5L BSVI RDE	**Gateway :** DR3	**Owner:** Ajaykumar Jadhav	**Project Name: SIGNA 2821.T 5L BSVI RDE(Cx Version)**	**Date:** 06/06/2023

Level 1 Summary:

	PALS Position	Status: Current/ Gateway	Status: Job#1 Predicted
Level 1:	L	L	

Statement Of Intent (SOI) or DNA Description:

Fuel Economy: Haulage Application

Attribute Target Summary:

Level 2 / Level 3 / Level 4 / Level 5 / Level 6	PALS	Metric	Target	Status: Current/Gateway SIGNA 2821.T Cx BSVI RDE	Status: Job#1 Predicted	Base Vehicle: Base Signa 2821.T BSVI	Remarks
Specification				--		--	
Vehicle GVW (Payload)		kg		28000		28000	
Power (Max) @ rpm		HP		204 hp @ 2200 rpm		200 hp @ 2200 rpm	
Power to Weight ratio		HP/Ton		7.29		7.14	
Torque (Max) @ rpm		Nm		850 Nm @ 1000-1600 rpm,		850 Nm @ 1000-1600 rpm	
Torque to Weight ratio		Nm/Ton		30.36		30.36	
Tyres size		--		Bridgestone 295/90R20		Apollo 295/90R20	
Gear Box Ratios		--		G950 DD, FGR -9.36		G950 DD, FGR -9.36	
Final Drive Ratio		--		5.12		4.88	
Fluid Economy at Rated Load	L			L			
Highway AC OFF		--					
Route : Jamshedpur-Ranchi		--					
Avg. Fluid Economy @40 kmph		kmpl	**≥ 4.45**	4.66		4.45	*Signa 2821.T OBII Cx and Signa 2821.T OBDI was tested back to back condition 0n JSR-Ranchi route.
Avg. Fluid Economy @55 kmph		kmpl	**≥ 4.05**	4.20		4.05	
Fluid Economy At Unladen	L			L			
Highway AC OFF		--					
Route : Jamshedpur-Ranchi		--					
Avg. Fluid Economy @40 kmph		kmpl	**≥ 7.86**	8.44		7.86	*Signa 2821.T OBII Cx and Signa 2821.T OBDI was tested back to back condition 0n JSR-Ranchi route.
Avg. Fluid Economy @55 kmph		kmpl	**≥ 6.78**	7.21		6.78	
Overall Fuel Economy (80% Laden + 20% Unladen)	L			L			
Highway AC OFF		--					
Route : Jamshedpur-Ranchi		--					
Overall Avg. Fluid Economy @40 kmph		kmpl	**≥ 4.87**	5.12		4.87	*Signa 2821.T OBII Cx and Signa 2821.T OBDI was tested back to back condition 0n JSR-Ranchi route.
Overall Avg. Fluid Economy @55 kmph		kmpl	**≥ 4.40**	4.59		4.40	

Executive Summary:

1. Test vehicle SIGNA 2821.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 15W50 engine oil + Low viscous 80W90 rear axle oil + Bridgestone LCRR tyres + 24" Fan
2. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 2821.T OBD-1 BSVI vehicle.
3. SIGNA 2821.T Cx RDE BSVI is meeting the PALS target of "Leadership".

Tested by:	Verified and Approved by:
Palash Das / Swapnil Jadhav **PAT ENGINEER**	Ajaykumar Jadhav / Sanjeev Kumar **PAT LEAD / PROJECT LEAD**

Annexure-2

TATA MOTORS LTD. ERC,PUNE	PAT ATTRIBUTES SHEET - Fuel Economy	VE CVBU/FE/01
		PAT Sheet No.: VE CVBU/ HCV/ SIGNA 2821.T 5L BSVI RDE/ Cx / FE/02

Depart ment Name: Vehicle Engineering

Project Name: SIGNA 2821.T 5L BSVI RDE_Cx (G-950 DD FGR-9.36, 5.12 RAR, 295/90R 20 phase II low CRR Tyres, 3 Mode FE Switch)
Base Vehicle: SIGNA 2821.T 5L BSVI OBD-1_G-950 DD, FGR-9.36, 4.88 RAR, 295/90R 20 Tyres_3 Mode FE Switch
Application: Haulage

Date: 27/06/2023

Sr.No.	Aggregate and test parameter	Unit	Test Type	Classification @DR4	Ref.Standard / Work Guideline	Simulation	Base Vehicle SIGNA 2821.T BSVI OBD-1 (G 950 DD | RAR 4.88 | 295/90R20)	Target	Test Vehicle SIGNA 2821.T BSVI OBD-II Cx (G 950 DD | RAR 5.12 | 295/90R20)	Status (RYG)	Remarks	CoC Remark
	Level 2 / Level 3 / Level 4 / Level 5 / Level 6 / Level 7											
A	**Fuel Economy Evaluation**			Heading			--		--	--		
2.0	Fuel Economy At Rated Load			Heading								
2.7	Highway AC OFF			Heading								
2.7.1	**Route : Jamshedpur-Ranchi**			Specification								
2.7.9	Average Fuel Economy @40 kmph	Kmpl		Data Genration			4.51	--	4.76	--		
2.7.10	Average Fuel Economy @55 kmph	Kmpl	Standard	Data Genration	TST/VT/WG/158		4.09	--	4.27	--	*Signa 2821.T OBD II Cx and Signa 2821.T OBD I was tested back to back condition on JSR-Ranchi route.	
2.7.11	**Average Fluid Economy @40 kmph**	Kmpl		Target			**4.45**	≥ 4.45	**4.66**	G		
2.7.12	**Average Fluid Economy @55 kmph**	Kmpl		Target			**4.05**	≥ 4.05	**4.20**	G		
4.0	Fuel Economy At Unladen			Heading								
4.7	Highway AC OFF			Heading								
4.7.1	**Route : Jamshedpur-Ranchi**			Specification								
4.7.9	Average Fuel Economy @40 kmph	Kmpl		Data Genration			8.01	--	8.59	--		
4.7.10	Average Fuel Economy @55 kmph	Kmpl	Standard	Data Genration	TST/VT/WG/158		6.86	--	7.27	--	*Signa 2821.T OBD II Cx and Signa 2821.T OBD I was tested back to back condition on JSR-Ranchi route.	
4.7.11	**Average Fluid Economy @40 kmph**	Kmpl		Target			**7.86**	≥ 7.86	**8.44**	G		
4.7.12	**Average Fluid Economy @55 kmph**	Kmpl		Target			**6.78**	≥ 6.78	**7.21**	G		
5.0	Overall Fuel Economy (80% Laden + 20% Unladen)			Heading								
5.1	Highway AC OFF			Heading								
5.1.1	**Route : Jamshedpur-Ranchi**			Specification								
5.1.9	Average Overall Fuel Economy @40 kmph	Kmpl		Data Genration			4.94	--	5.23	--		
5.1.10	Average Overall Fuel Economy @55 kmph	Kmpl		Data Genration	TST/VT/WG/158		4.45	--	4.65	--	*Signa 2821.T OBD II Cx and Signa 2821.T OBD I was tested back to back condition on JSR-Ranchi route.	
5.1.11	**Overall Average Fluid Economy @40 kmph**	Kmpl		Target			**4.87**	≥ 4.87	**5.12**	G		
5.1.12	**Overall Average Fluid Economy @55 kmph**	Kmpl		Target			**4.40**	≥ 4.40	**4.59**	G		

Sign Off			Index for Status
Responsibility	**Name**	**Signature**	Off target, no resolution
PAT Engineer	**Palash Das / Swapnil Jadhav**		Marginally off target, resolution planned/low risk
PAT Lead / Project Lead	**Ajaykumar Jadhav / Sanjeev kumar**		Met target, no concern

Fuel Economy Evaluation on Signa 2821.T 5L RDE BSVI with Signa 2821.T 5L BSVI Base Vehicle.

PED - HCV I 27.06.2023

OUR CULTURE CREDO

AT TATA MOTORS
We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.
We are **MORE WHEN ONE!**

BE BOLD
Taking calculated **risk** is key to making progress. We act with confidence and **agility** to accomplish our goals

OWN IT
Feeling and acting **empowered** is critical to drive results. We have an **Owner's Mind-set** and each of us takes full responsibility for the outcomes

SOLVE TOGETHER
Leveraging our collective genius while holding each other **accountable** helps us deliver the best. We **collaborate** proactively and transparently to achieve innovative solutions

BE EMPATHETIC
Embracing diversity makes us stronger for differences are opportunities to learn. We work with **passion to delight customers** and deliver greater success to our stakeholders

Vehicle Specification

TATA MOTORS
Connecting Aspirations

Specification		Test Vehicle	Base Vehicle
Vehicle	Model	Signa 2821.T TML5L BSVI RDE	Signa 2821.T TML5L BSVI
	GVW (Rated load) Kg	28000(Kg)	28000(kg)
	GVW (Unladen) kg	--	--
Engine	Emission norm	BSVI RDE	BSVI
	Engine	TML 5LNGE	TML 5LNGE
	Calibration	95%	As per production released
	Technology	--	--
	Torque	850 Nm @ 1000-1600 rpm	850 Nm @ 1000 - 1600 rpm
	Power	204 HP @ 2200 rpm	200 HP @ 2200 rpm
	Torque to weight ratio	30.35	30.35
	Power to weight ratio	7.28	7.14
	Torque band	600	600
Driveline	Gear box	G950 6S DD, FGR- 9.36	G950 6S DD, FGR- 9.36
	Gear 1	9.36	9.36
	Gear 2	5.2	5.2
	Gear 3	3.05	3.05
	Gear 4	1.84	1.84
	Gear 5	1.33	1.33
	Gear 6	1	1
	RAR	5.12	4.88
	Tyre	295/90R20	295/90R20

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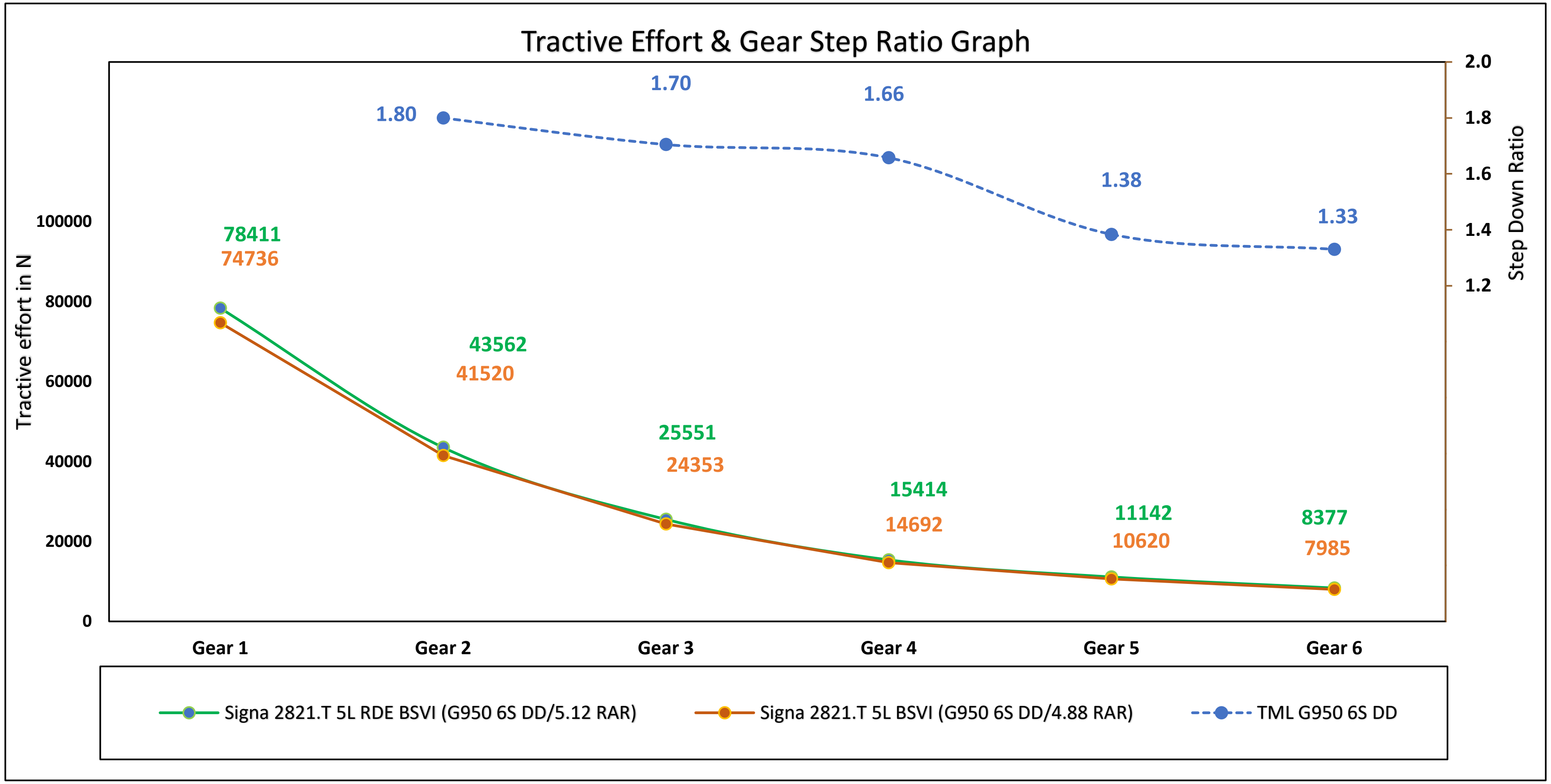
Tractive Effort Comparison

TATA MOTORS
Connecting Aspirations



- Tractive effort of Signa 2821.T RDE vehicle is better than Signa 2821.T BSVI vehicle.

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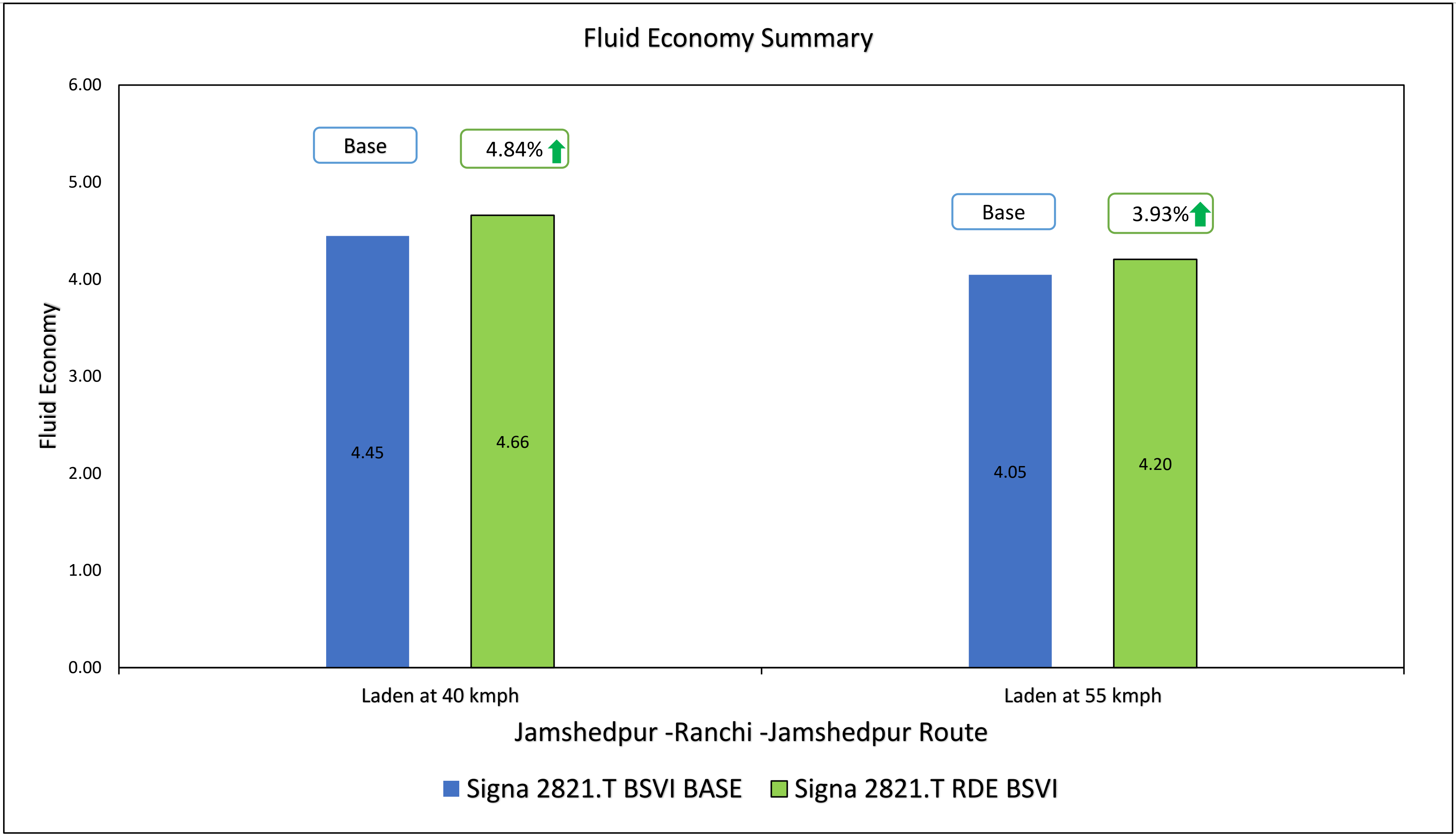
Summary FE Status with all FE Avenues(Laden)

TATA MOTORS
Connecting Aspirations



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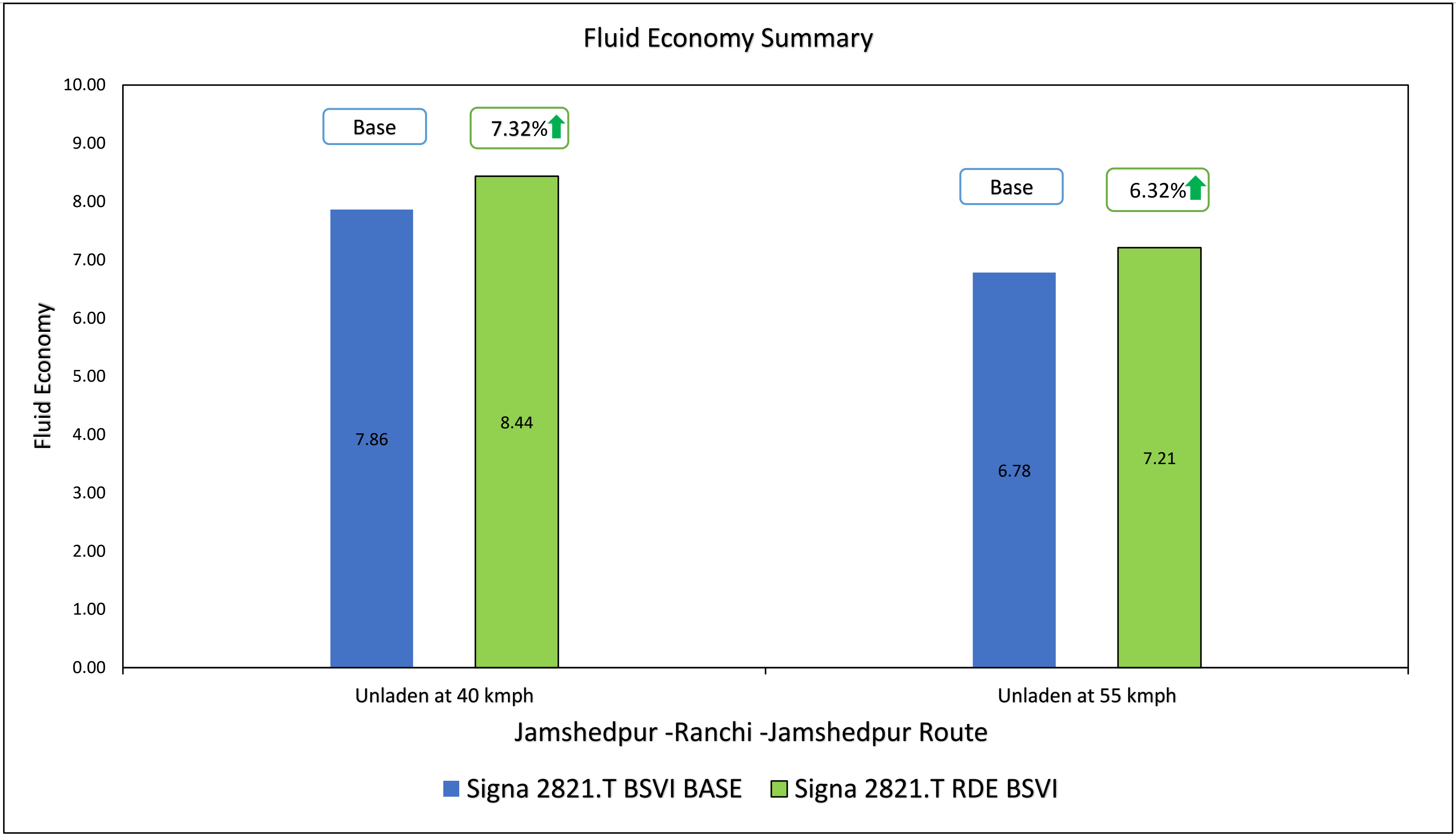
Summary FE Status with all FE Avenues(Unladen)

TATA MOTORS
Connecting Aspirations



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Duty Cycle Analysis

Jamshedpur-Ranchi Road FE Trial

Duty Cycle : Summary (Laden)

TATA MOTORS
Connecting Aspirations

JSR-RANCHI-JSR Duty Cycle

Parameters		05.05.2023(40 kmph) Signa 2821.T RDE Test Vehicle	04.05.2023(55 kmph) Signa 2821.T RDE Test Vehicle
Engine idle speed (rpm)		750	750
Low idle time (%)		--	--
Average engine speed (rpm)		1095.9	1,326.1
Average power (kW)		32.7	45.36
Avg. vehicle speed (kmph)	Including Stop	35.67	44.18
	Excluding Stop	--	--
FE switch utilization (%time)	Power	4.6	7.7
	Balance	11.7	11.2
	Eco	83.6	81.2
Gear utilization (%time)	Neutral Gear	3.0	3.6
	4th Gear	4.2	4.4
	5th Gear	11.0	6.4
	6th Gear	76.9	80.3
	Total shifts	--	--
Throttle (%time)	100%	20.4	30.4
	0%	36.4	31.8
Coolant Temp (°C)	80°C	19.7	12.2
	85°C	22.2	14.1
	90°C	18.3	29.6
SCRTM Fuel spent (%)		--	--
Regeneration Fuel spent (%)		--	--
Duty Cycle fuel economy (kmpl)		4.73	4.2
Duty cycle Distance(km)		227	228
Duty Cycle Time(Hrs)		5.84	5.25
Actual Avg. Fuel Economy		4.76	4.27

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Duty Cycle : Summary (Unladen)

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JSR-RANCHI-JSR Duty Cycle

Parameters		14.05.2023(40 kmph) Signa 2821.T RDE Test Vehicle	11.05.2023(55 kmph) Signa 2821.T RDE Test Vehicle
Engine idle speed (rpm)		**750**	**750**
Low idle time (%)		--	--
Average engine speed (rpm)		1007.5	1271.9
Average power (kW)		15.66	23.67
Avg. vehicle speed (kmph)	Including Stop	34.58	43.13
	Excluding Stop	--	--
FE switch utilization (%time)	Power	0.00	0.00
	Balance	1.6	0.6
	Eco	98.3	99.3
Gear utilization (%time)	Neutral Gear	4.6	6.9
	4th Gear	1.3	1.7
	5th Gear	1.2	2.7
	6th Gear	89.4	84.2
	Total shifts	--	--
Throttle (%time)	100%	0.6	1.2
	0%	26.7	30.8
Coolant Temp (°C)	80°C	**3.8**	**28.8**
	85°C	**1.6**	**31.5**
	90°C	**0.8**	**19.2**
SCRTM Fuel spent (%)		--	--
Regeneration Fuel spent (%)		0.00	0.00
Duty cycle fuel economy (kmpl)		**8.15**	**7.19**
Duty cycle Distance(km)		**224**	**225**
Duty Cycle Time(Hrs)		**5.84**	**5.25**
Actual Avg. Fuel Economy		**8.59**	**7.27**

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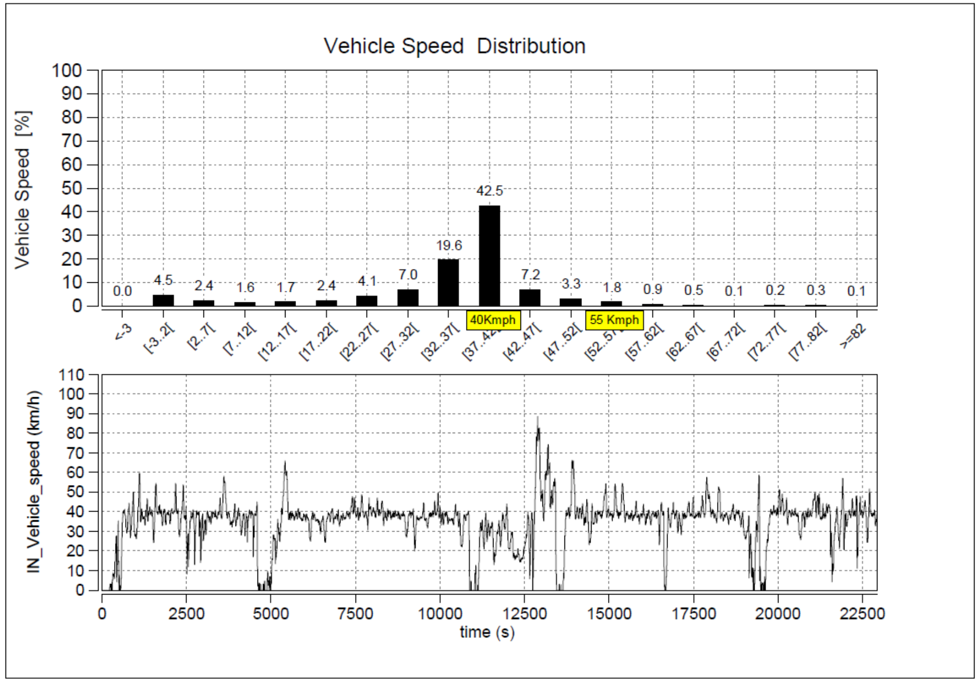
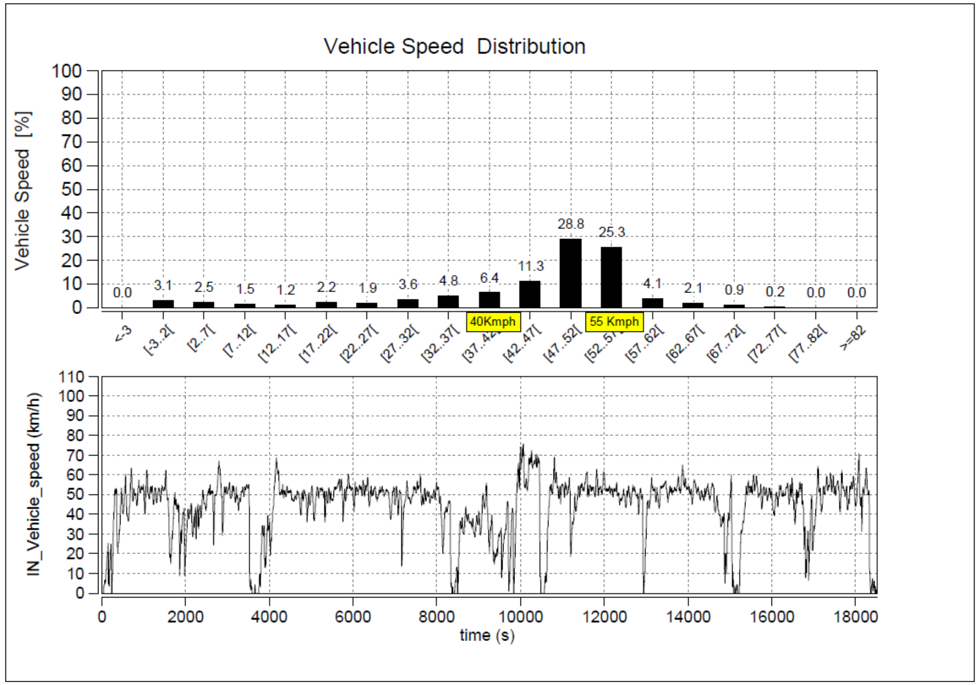
Vehicle Speed Distribution

Laden Trial

Signa 2821.T RDE Test Vehicle

05.05.23	40 kmph	04.05.23	55 kmph



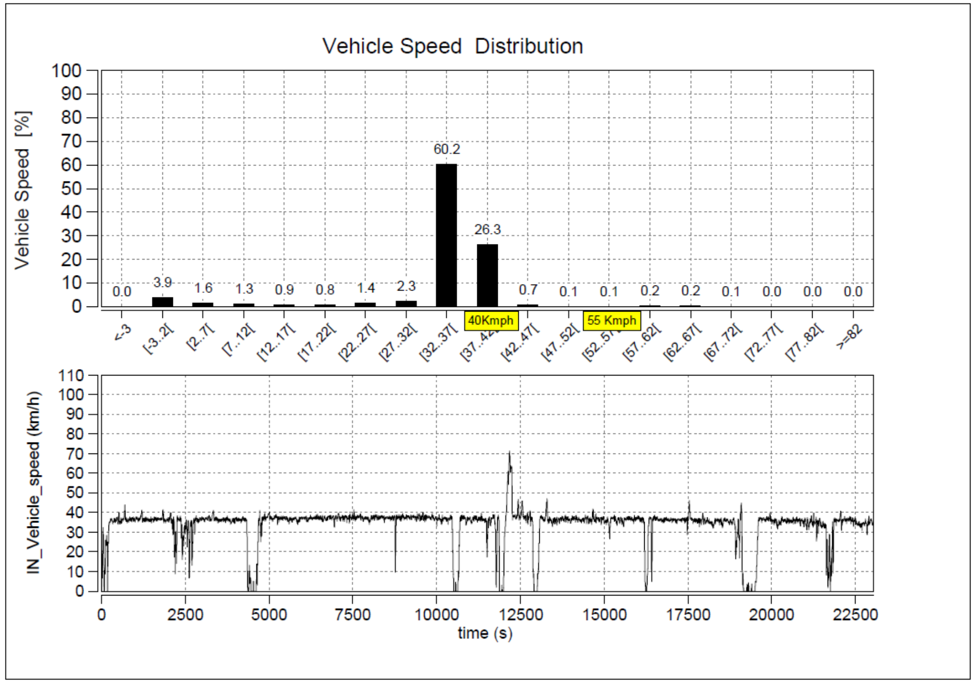


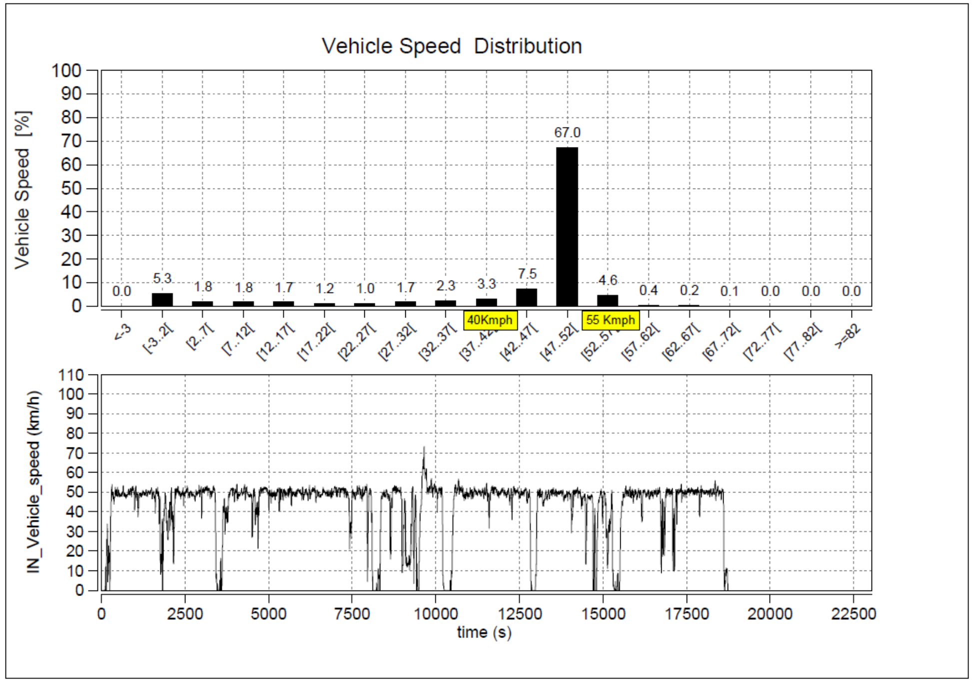
Vehicle Speed Distribution

Unladen Trial

Signa 2821.T RDE Test Vehicle

14.05.23	40 kmph	11.05.23	55 kmph





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Engine Speed Distribution

Laden Trial

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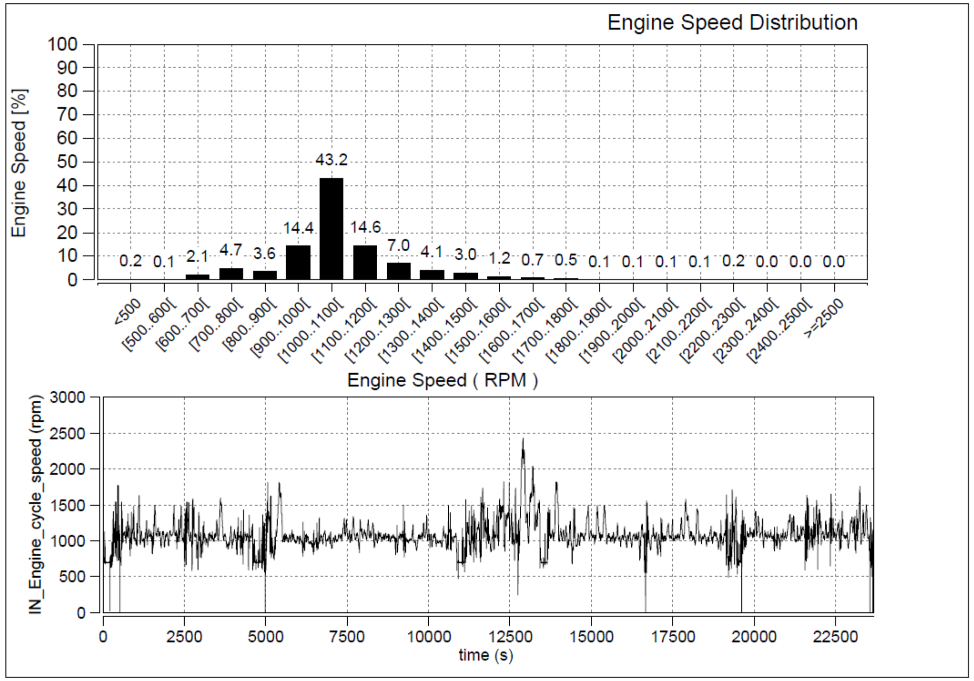
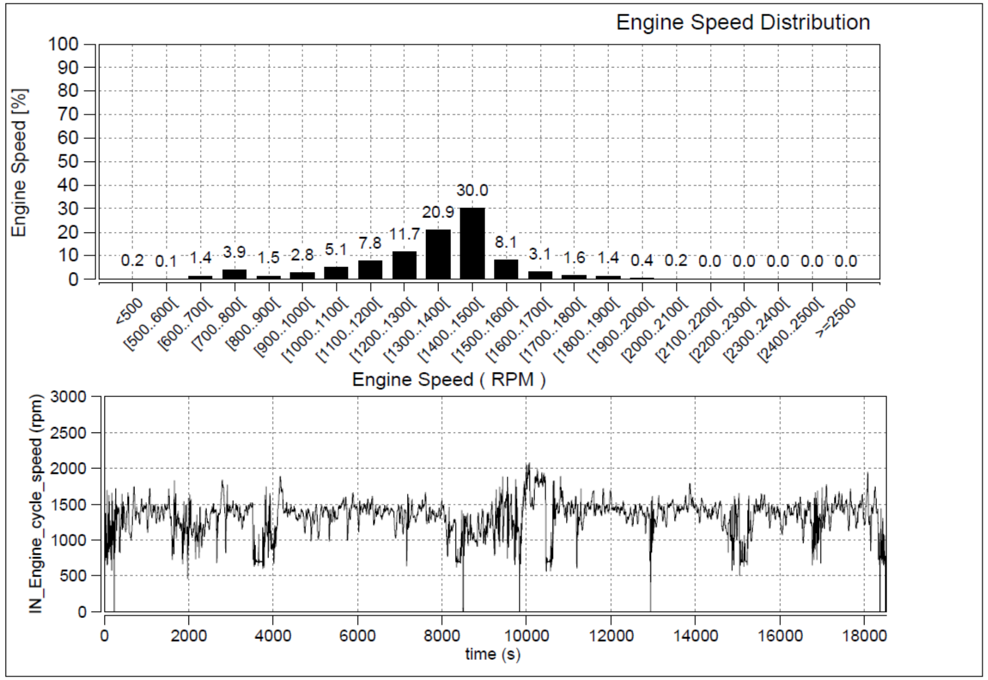
Signa 2821.T RDE Test Vehicle

05.05.23	40 kmph	04.05.23	55 kmph





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Engine Speed Distribution

Unladen Trial

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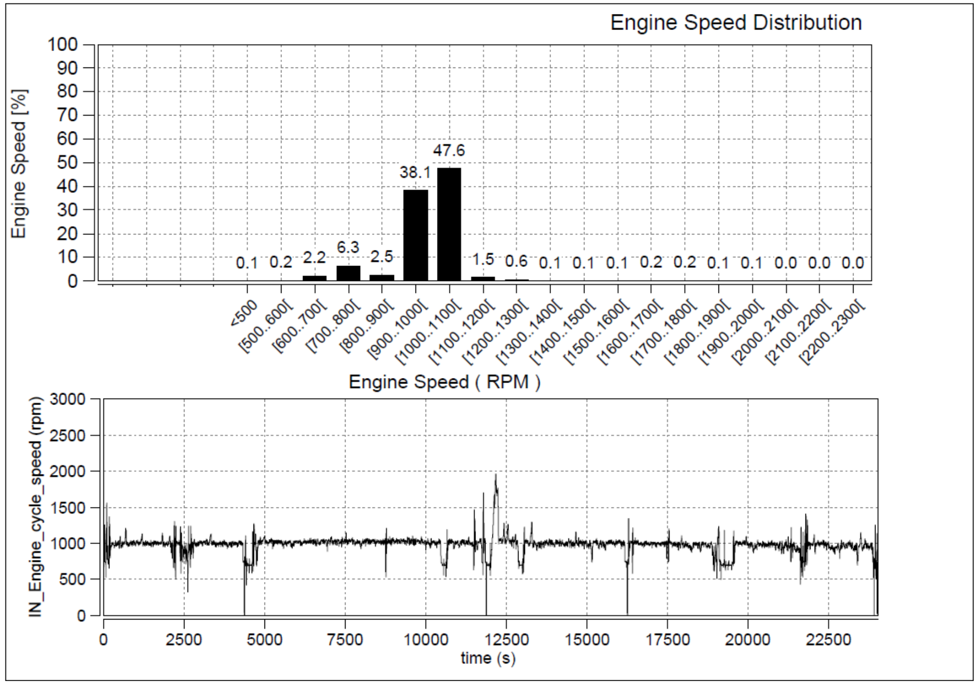
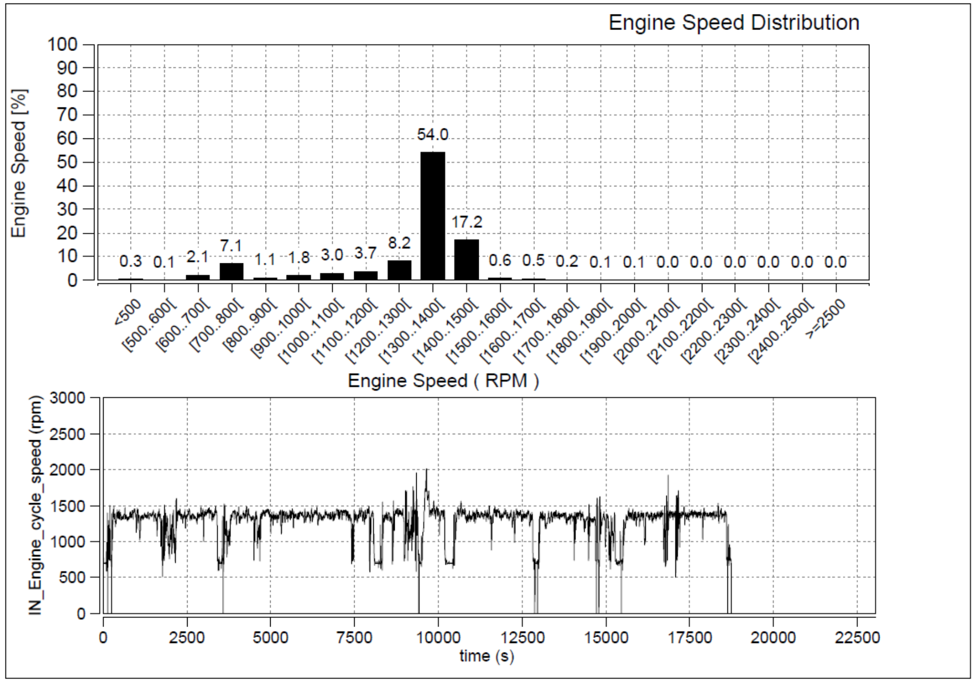
Signa 2821.T RDE Test Vehicle

14.05.23 **40 kmph**



11.05.23 **55 kmph**



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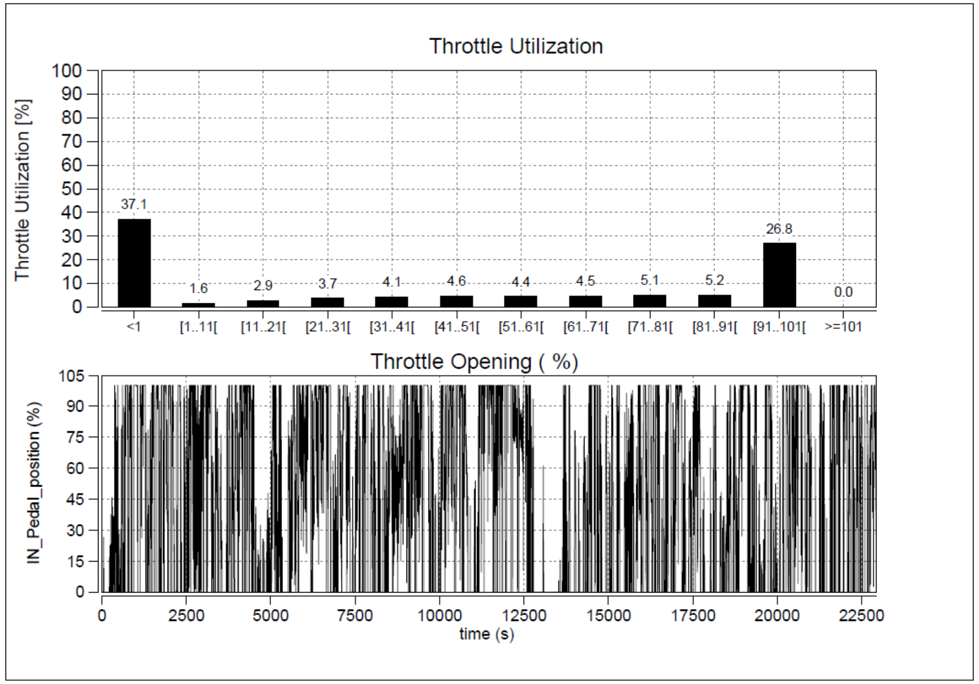
Throttle Distribution

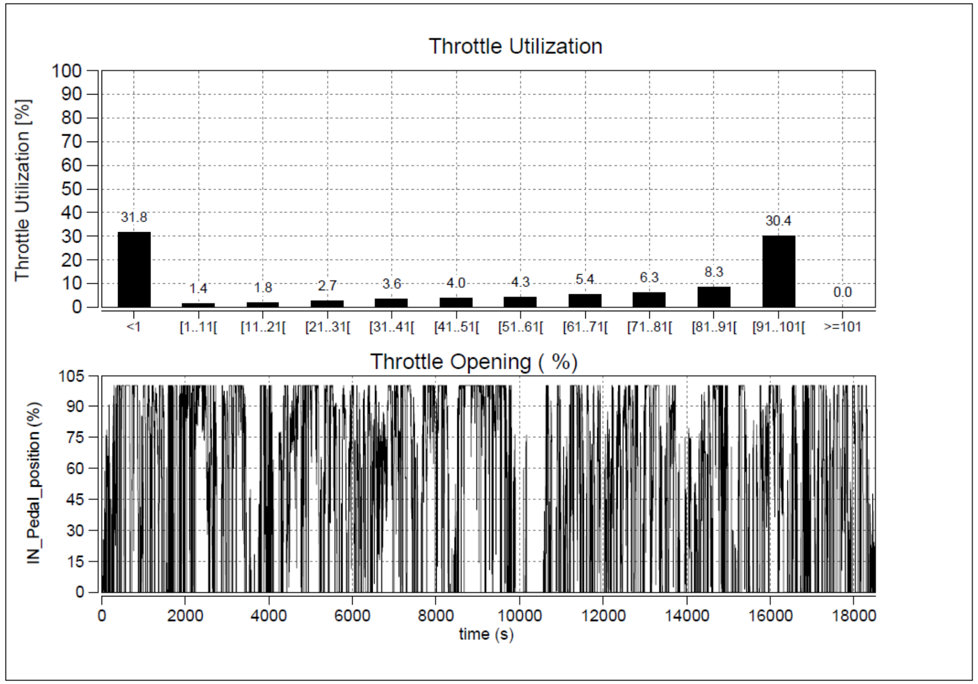
Laden Trial

Signa 2821.T RDE Test Vehicle

05.05.23 **40 kmph**



04.05.23 **55 kmph**

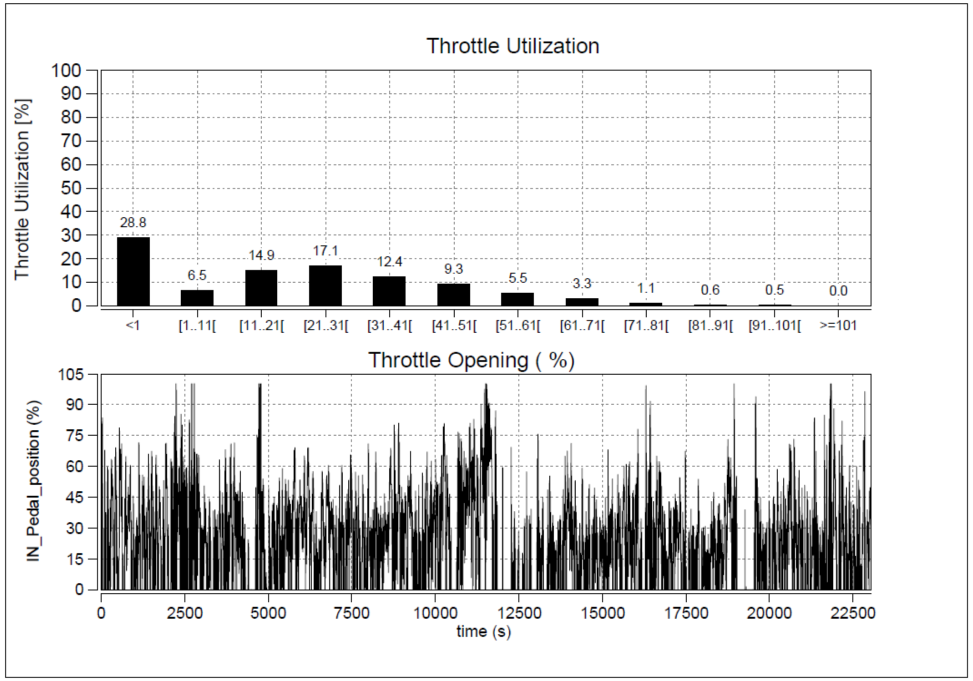


Throttle Distribution

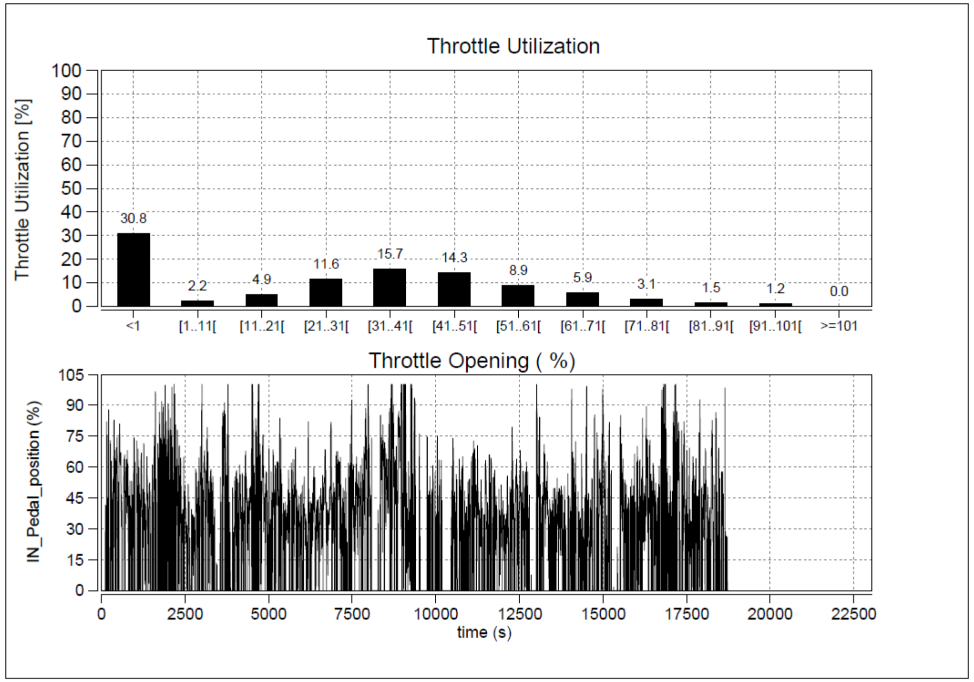
Unladen Trial

Signa 2821.T RDE Test Vehicle

14.05.23 **40 kmph**



11.05.23 **55 kmph**

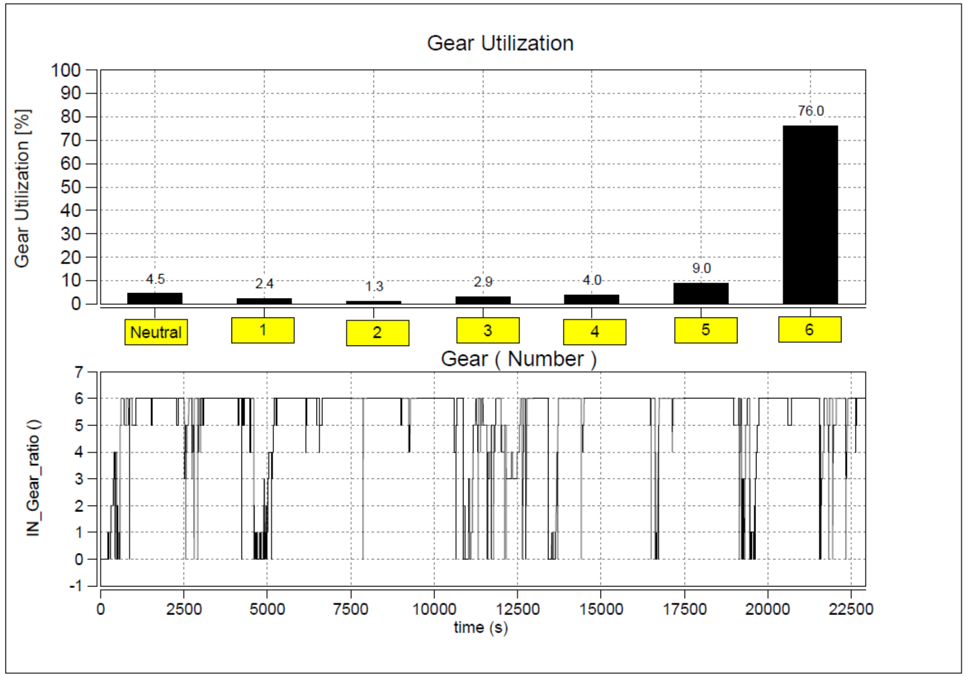


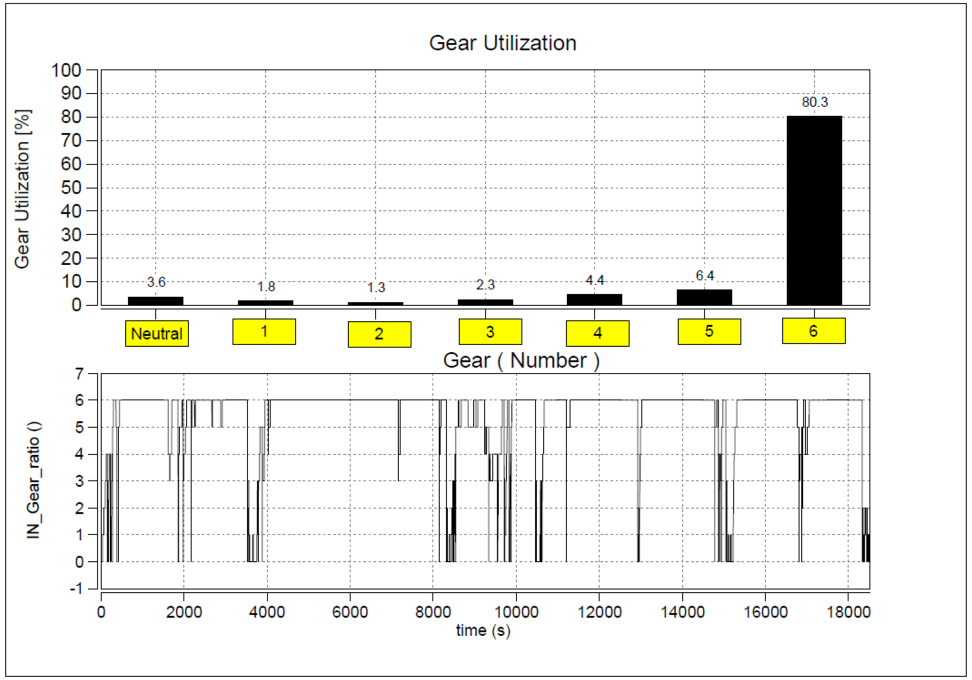
Gear Utilization

Laden Trial

Signa 2821.T RDE Test Vehicle

05.05.23	40 kmph	04.05.23	55 kmph



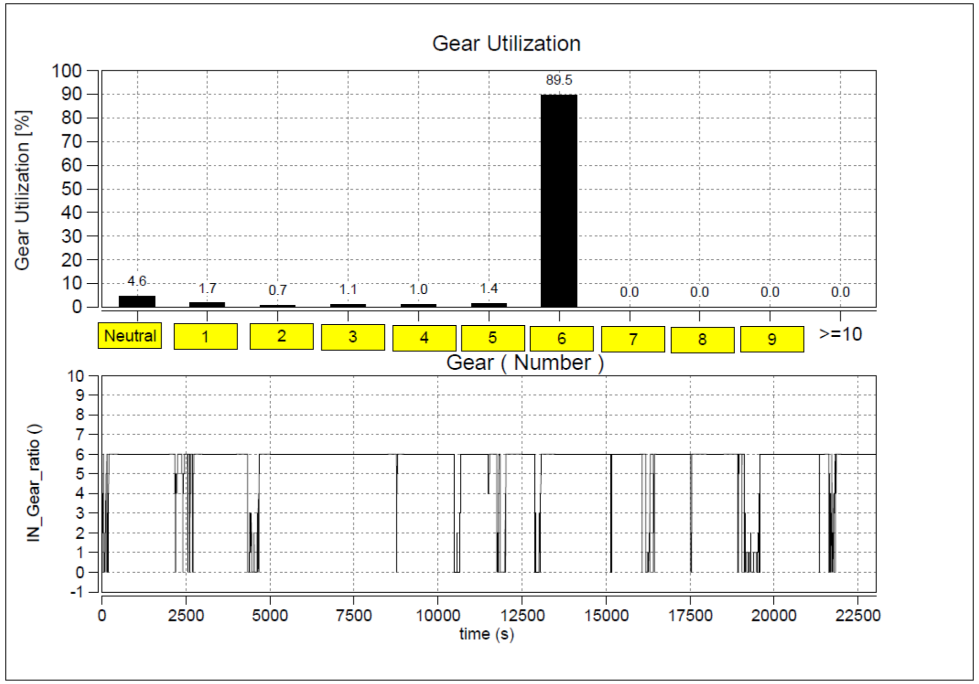


Gear Utilization

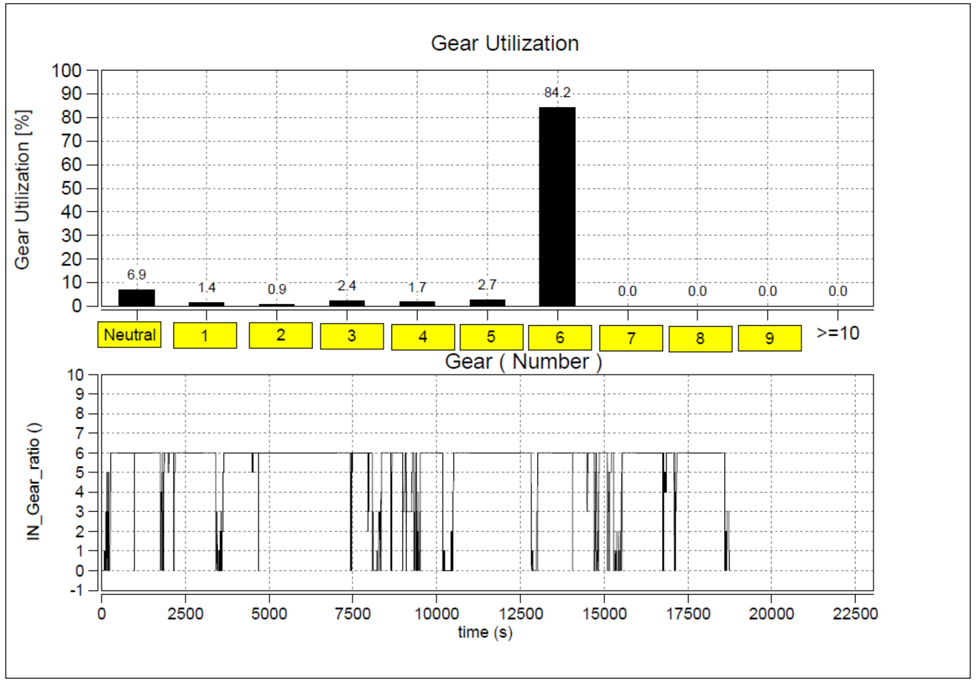
Unladen Trial

Signa 2821.T RDE Test Vehicle

14.05.23 **40 kmph**



11.05.23 **55 kmph**



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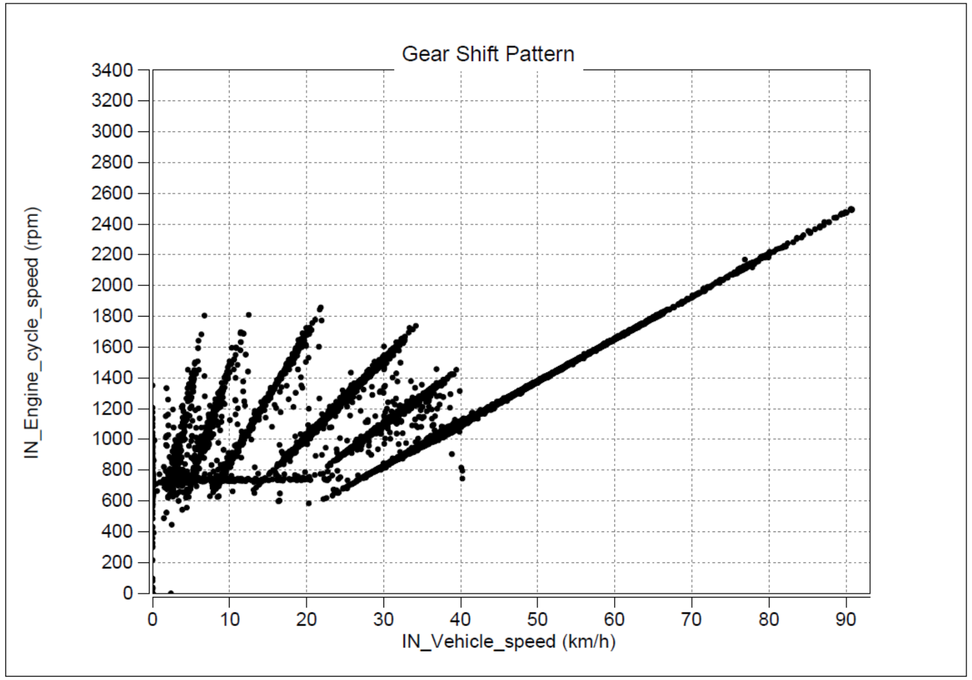
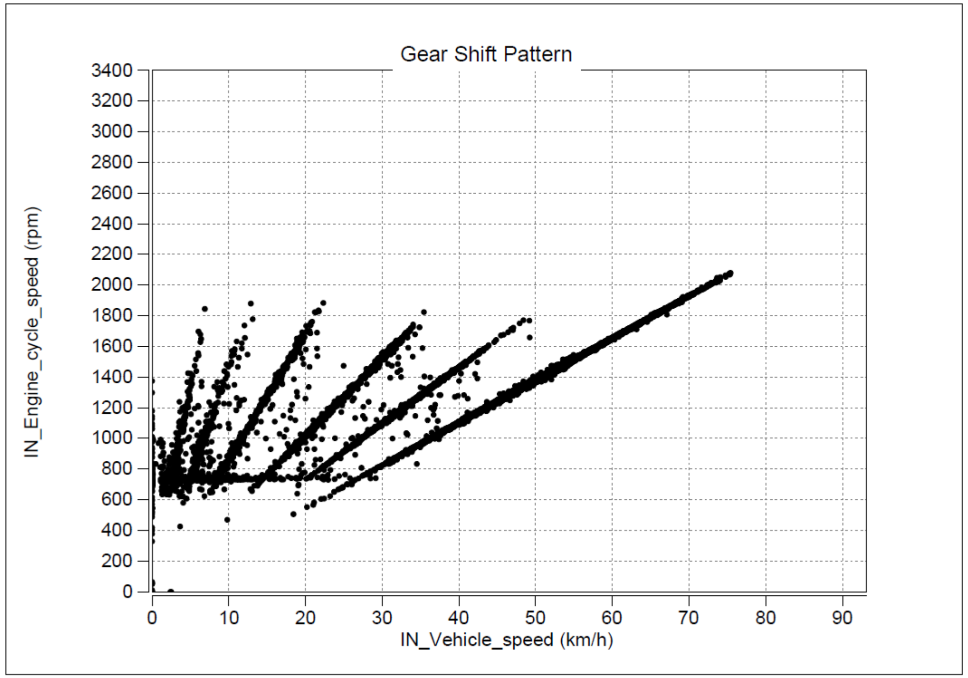
Transmission Shift Pattern

Laden Trial

Signa 2821.T RDE Test Vehicle

05.05.23	40 kmph	04.05.23	55 kmph





Transmission Shift Pattern

Unladen Trial

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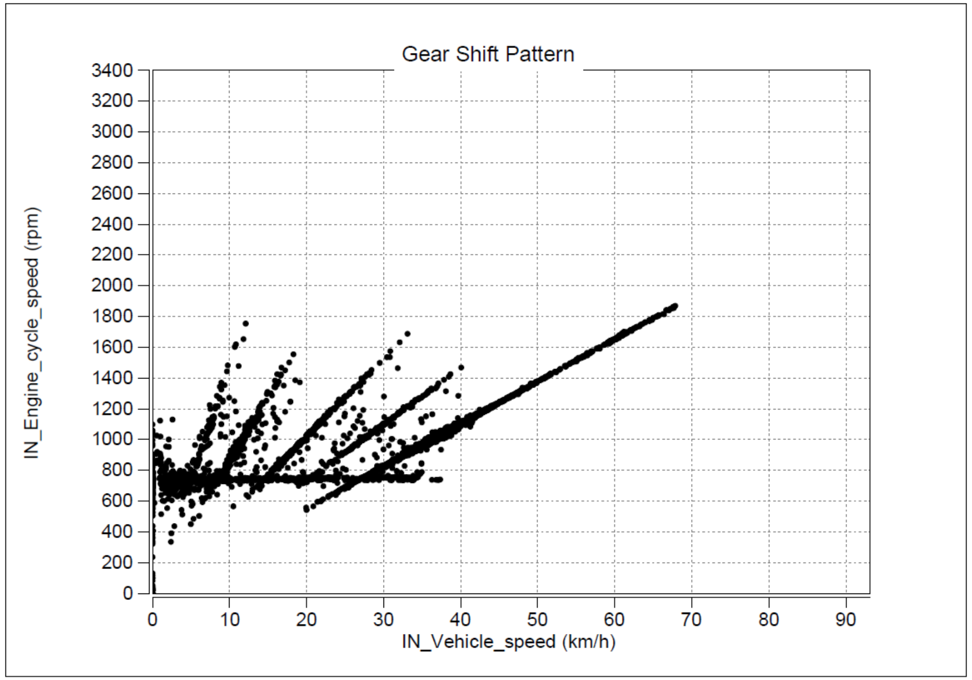
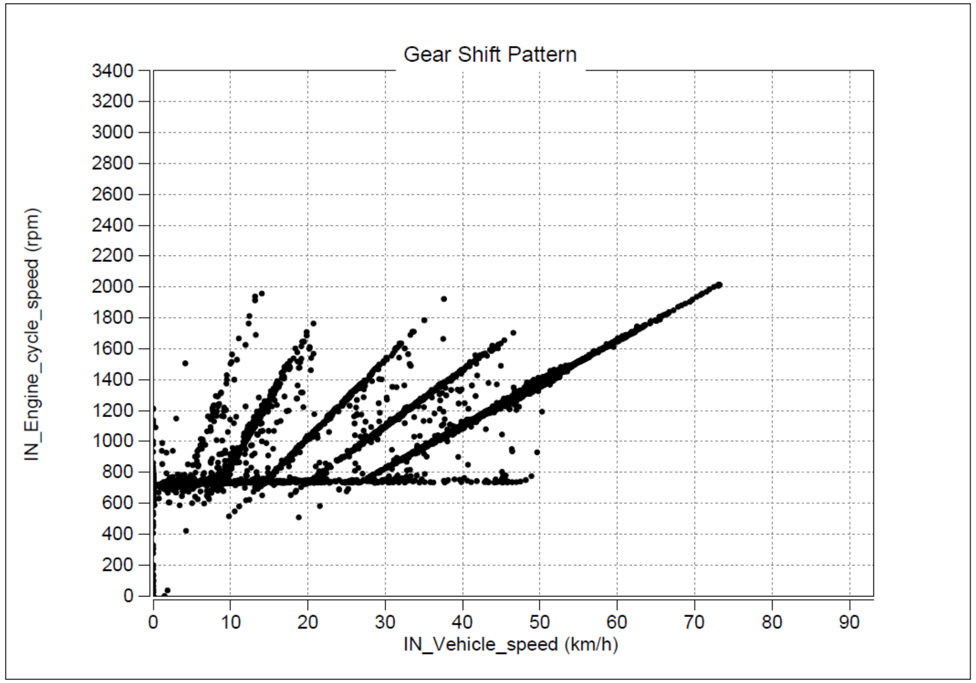
Signa 2821.T RDE Test Vehicle

14.05.23 **40 kmph**



11.05.23 **55 kmph**



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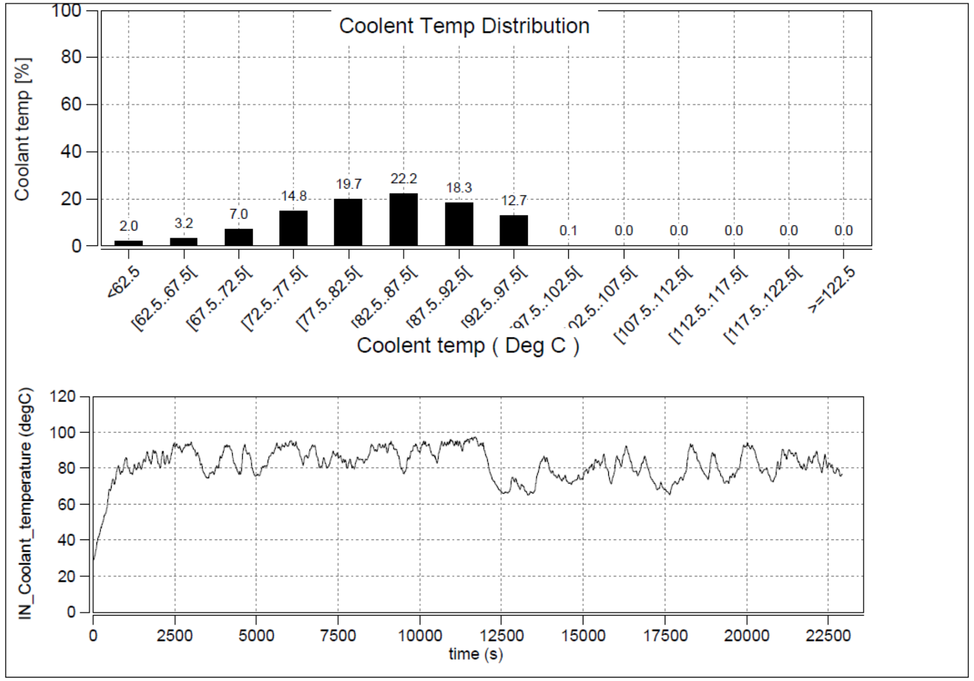
Coolant Temp Distribution

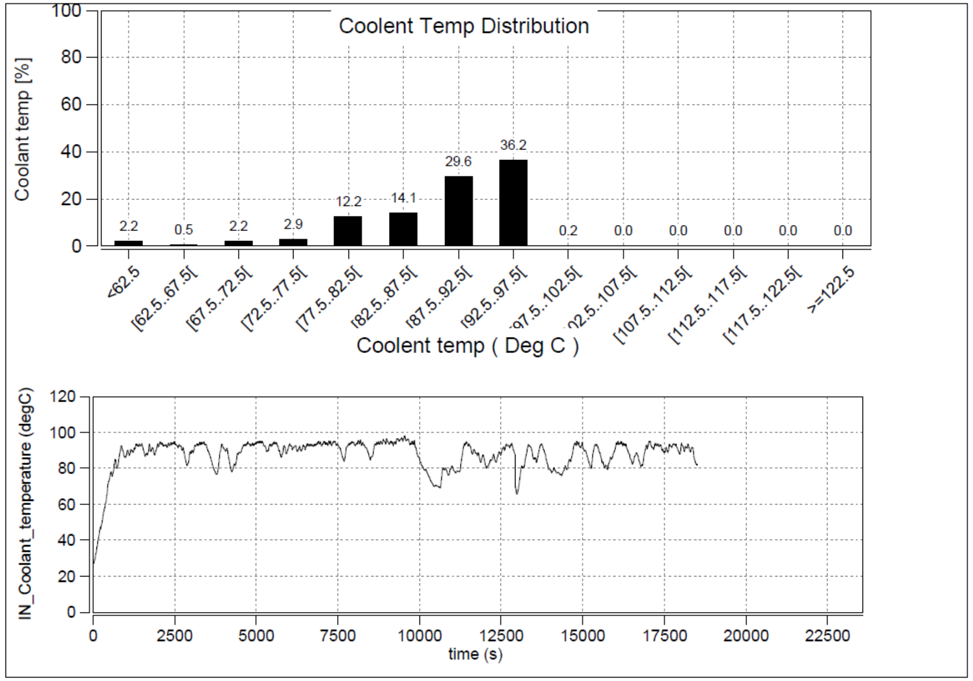
Laden Trial

Signa 2821.T RDE Test Vehicle

05.05.23	40 kmph



04.05.23	55 kmph

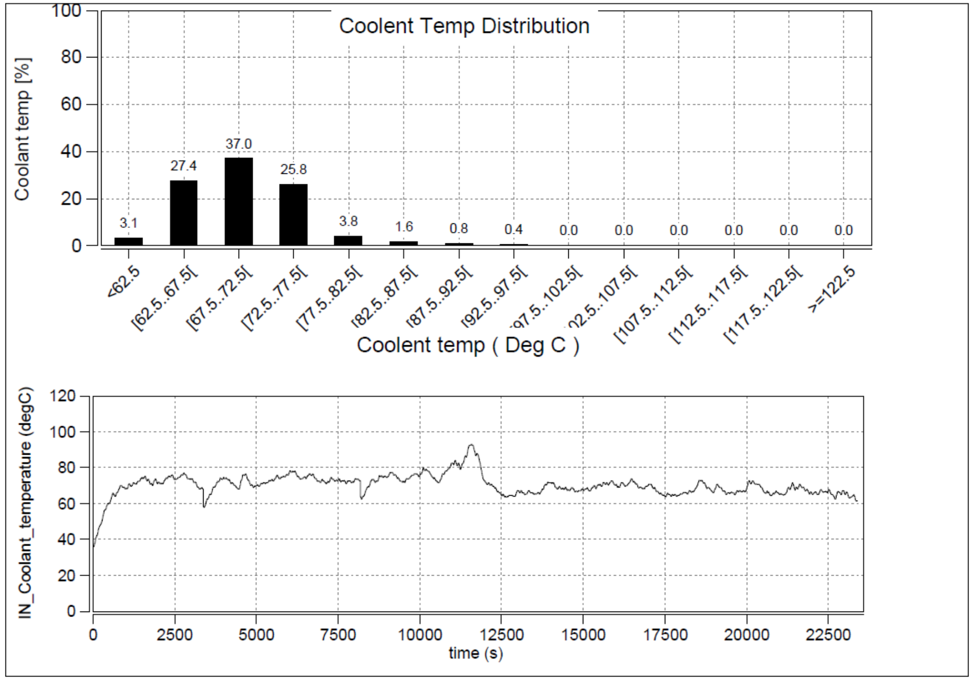


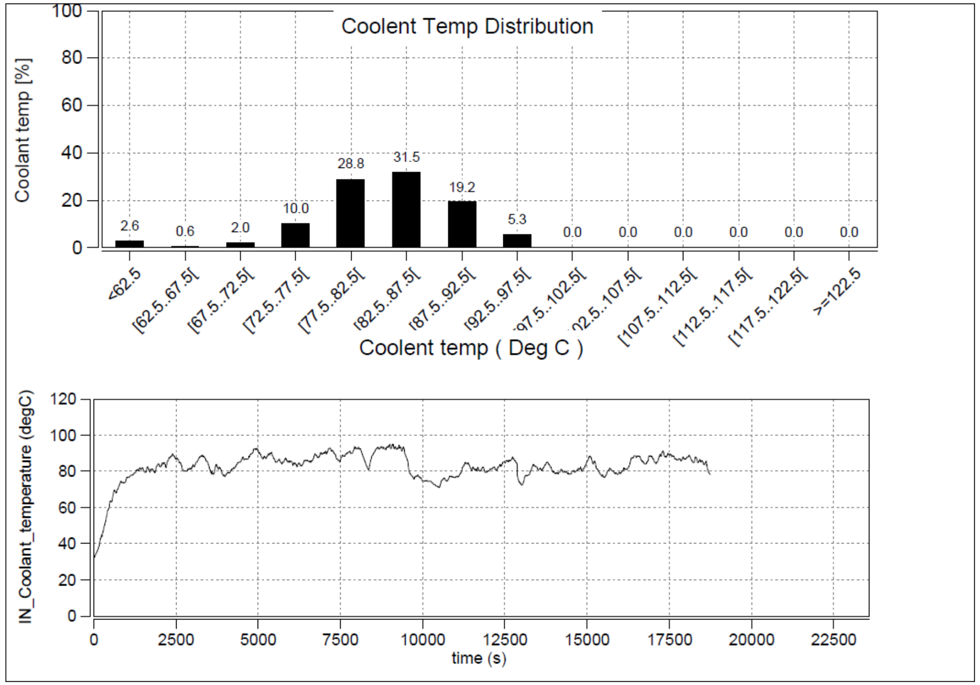
Coolant Temp Distribution

Unladen Trial

Signa 2821.T RDE Test Vehicle

14.05.23	40 kmph	11.05.23	55 kmph





Inference

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Fluid Economy(FLE) Status of 2821.T BSVI RDE Test vehicle w.r.t. 2821.T BSVI **base vehicle :**

1. Laden

a) 40kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by 4.84 % .

b) 55kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by 3.93 % .

2. Unladen

a) 40kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by 7.32 %.

b) 55kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by 6.32 %.

Overall Average Fluid Economy(FLE) Status of 2821.T BSVI RDE Test vehicle w.r.t. 2821.T BSVI base vehicle :

a) **40kmph (Laden(80%) +Unladen(20%)):-** Fluid Economy of 2821.T is **superior** than base vehicle by **5.14 % .**

b) **55kmph (Laden(80%) +Unladen(20%)):- :-** Fluid Economy of 2821.T is **superior** than base vehicle by **4.23 %** .

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Thank You

TEST REQUISITION

Ref.		Date:	31/03/2023

To,
ERC Testing
Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 2821.T 5L BSVI					
2	Part Number	51695248R,					
3	Drawing number	516900000003,	Mod status	B,2,	Mod Date	25/02/2023	
4	Vehicle Model / Project	TATA SIGNA 2821.T RDe BSVI					
5	WBS Number	P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T					
6	Supplier	NA					
7	Number of samples						
8	Inspection report	Enclosed	☑ Yes	☐ No	If No commitment date:		
9	Material report	Enclosed	☐ Yes	☑ No	If No commitment date:		
10	Suppliers test report	Enclosed	☑ Yes	☐ No	If No commitment date:		
11	Reason for testing	First Source Development	☐	Weight Reduction	☐		
		Alternate source development	☐	Cost Reduction	☐		
		Data generation Test	☐	Regulatory test	☐		
		Any other (pl. specify)	☐	New Development	☐		
12	What testing required?	DYNAMIC					
13	Remarks						
14	Peripheral Parts Availability	YES					

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN VEHICLE TESTING DEPARTMENT				**DATE:** 09/06/2023
Description: Signa 2821.T 5L RDE BSVI Cx		**Test Item No:** 42163			
		New	✔	**Moderate**	x
		Major	x	**Minor**	x

Part No: 51695248R, **Drawing No:** 516900000003,	**Modification No:** B,2,	**Enclosures:** Back to back FE Trials of Signa 2821.T 5L RDE BSVI Cx Vehicle
Make: NA	**WBS Number:** P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T	**Copies to** :
Vehicle Type: TATA SIGNA 2821.T RDe BSVI	**No. of Samples:** 1	
Test Requested by:	VEHICLE-I	**Equipments:** Component & System Level Performance Tests
Test Requisition Reference:	Testing-Outdoor/15049	
Reason for Test/s:	Component & System Level Performance Tests	
Regulatory Requirement:	a) Component:	
	b) Vehicle:	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	FE Trials of Signa 2821.T 5L RDE BSVI Cx Vehicle	TST/VT/WG/158	As per PALS	1	Palash Das	Beta	completed

REMARKS: FE trial	**INTERNAL REF:**	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	**CHECKED BY**	**APPROVED BY**